

Exact Cycle Classification for Every Finite Kac Ring

Route-A structural certificate C170

Abstract

For every ring size and every binary marker configuration, we classify all cycles of the reversible finite Kac ring. The marker product η is the only invariant needed: positive product gives two cycles of length N , while negative product gives one cycle of length $2N$. We derive all fixed counts, the Artin–Mazur zeta, and the finite Koopman determinant.

Keywords: Kac ring; marker configuration; exact period; primitive cycle; dynamical zeta; Koopman permutation; time reversal.

中文摘要

本文对任意环长和任意二元标记配置的可逆有限Kac环给出完整周期分类。所有动力信息只依赖标记乘积 η ：正乘积产生两个长度为 N 的周期，负乘积产生一个长度为 $2N$ 的周期。由此严格导出所有时刻的不动点数、动力 ζ 函数、有限维Koopman行列式与反酉时间反演。

关键词：Kac环；标记配置；精确周期；本原周期；动力 ζ 函数；Koopman置换；时间反演。

1 All-marker classification

On $X_N = \mathbb{Z}/N\mathbb{Z} \times \{\pm 1\}$ let

$$T(j, s) = (j + 1, \varepsilon_j s), \quad \eta = \prod_{j=0}^{N-1} \varepsilon_j.$$

One circuit gives $T^N(j, s) = (j, \eta s)$. Since every return time is divisible by N , all states have exact period N when $\eta = +1$ and exact period $2N$ when $\eta = -1$. Hence there are respectively two and one geometric cycles. With $L = N$ or $2N$ and $c = 2N/L$,

$$\# \text{Fix}(T^n) = 2N \mathbf{1}_{L|n}, \quad \zeta_T(z) = (1 - z^L)^{-c}, \quad \det(I - zU_T) = (1 - z^L)^c.$$

The eigenvalues of the counting-measure Koopman permutation are all L -th roots of unity, each with multiplicity c . The zeta exponent counts geometric cycles, whereas the fixed count uses all $2N$ labelled states; keeping those multiplicities separate prevents a factor-of- L error. The two $N = 1$ cases are included: positive product is the identity on two states, and negative product is one transposition.

2 Gauge and reversal

Set $g_0 = 1$, $g_j = \prod_{r < j} \varepsilon_r$, and $q = g_j s$. In (j, q) coordinates the map advances the site, preserves q away from the wrap, and sends q to ηq at the wrap. If $\eta = +1$, reflection $j \mapsto -j$ reverses each cycle. If $\eta = -1$, define $\psi(j, +1) = j$ and $\psi(j, -1) = N + j$; the map becomes translation by one on $\mathbb{Z}/(2N)\mathbb{Z}$, reversed by $t \mapsto -t$. Pullback through the gauge yields an involution R with $RTR = T^{-1}$. Thus $\Theta f = f \circ R$ is antiunitary and $\Theta U_T \Theta = U_T^{-1}$. The permutation unitary is self-adjoint exactly when $L \leq 2$.

3 Route-A decision and evidence boundary

Gate	Verdict	Reason
A0	FAIL	no intrinsic arithmetic origin
A1	WEAK	complete but finite reducible cycles
A2	FAIL	no target divisor comparison
A3	FAIL	no target global comparison
A4	NATURAL_QUANTIZATION	same-clock permutation and reversal

The v0.2 overall verdict is `ROUTE_A_REJECTED`, because A0 fails; Route B is false. The paper is retained as theorem progress, not as a primary Hilbert–Pólya candidate.

The release ledger contains 48 class rows through $N = 24$. An independent implementation exhausts all 2,046 marker words through $N = 10$, covering 36,868 labelled states, 36,868 fixed-time checks, and 73,736 reversal identities. Its checker passes 114,056 assertions, SymPy passes 221 matrix and polynomial checks, byte replay is exact, and 17 hostile mutations are rejected. These are regression sentinels only; the all-size, all-marker result rests on the circuit and gauge proofs. Citation and reference registries each have population zero, and no novelty or priority claim is made.

Limitations. The system is a finite, exactly reducible kinetic toy model. Its primitive cycles have no proved prime semantics. We make no target divisor, functional-equation, counting-law, arithmetic local-factor, Euler-factor, root-number, automorphy, or Hilbert–Pólya claim. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript.

Ethics. No human, animal, clinical, personal, or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

Funding. No external funding is reported.

Conflicts of interest. None known.

AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.